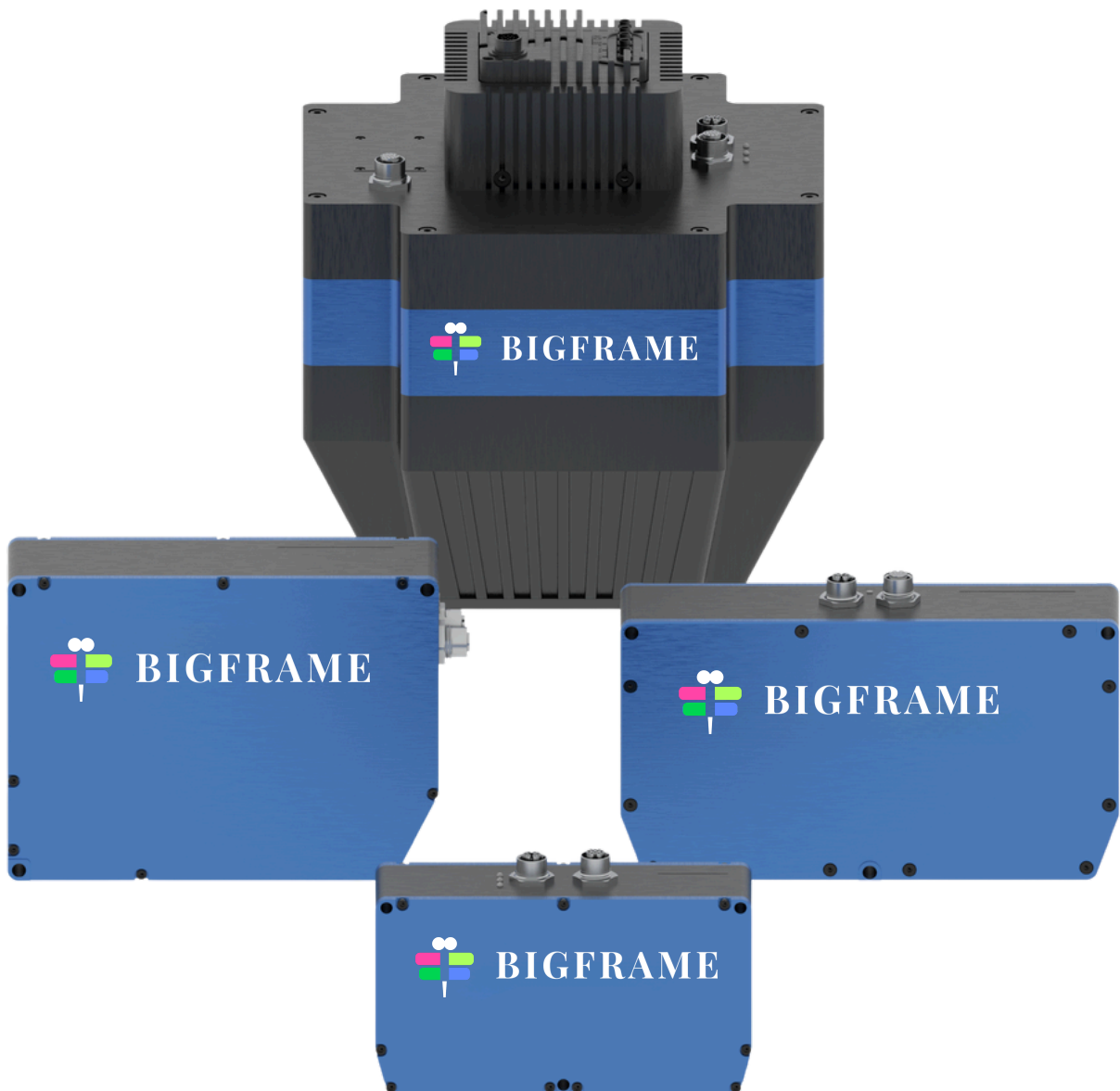




BIGFRAME

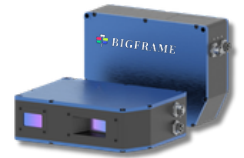


BEYOND VISION

Area-scanning Structured Light 3D Camera

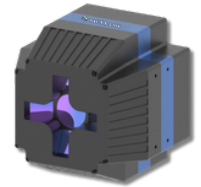
VRH9 Series

- 9.07 MP 3D point cloud
- Field of view spans 24 mm to 302 mm
- Z-axis repeatability up to 0.15 μm
- High-precision measurement for semiconductor, 3C electronics, automotive parts, and lithium-battery production



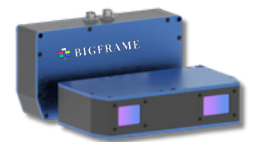
TRS Series

- 26 MP 3D point cloud
- Four-projector design delivers full coverage with no blind spots, Field of view from 9.8 mm to 25.5 mm
- Z-axis repeatability up to 0.06 μm
- Fine inspection of very small parts in the semiconductor industry and beyond



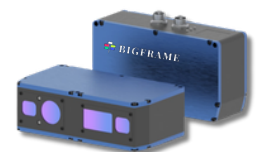
VR Series

- 3.2 MP 3D point cloud
- Field of view spans 215 mm to 2180 mm
- Z-axis repeatability up to 2 μm
- Built for guidance, positioning, and 3D inspection in medium-to-large scenes



VRD Series

- 5 MP 3D point cloud with RGB image output
- Binocular 3D imaging, Field of view spans 365 mm to 1538 mm
- Z-axis repeatability up to 6 μm
- Lightweight and compact — made for "eye-on-hand" 3D guidance on collaborative robot arms

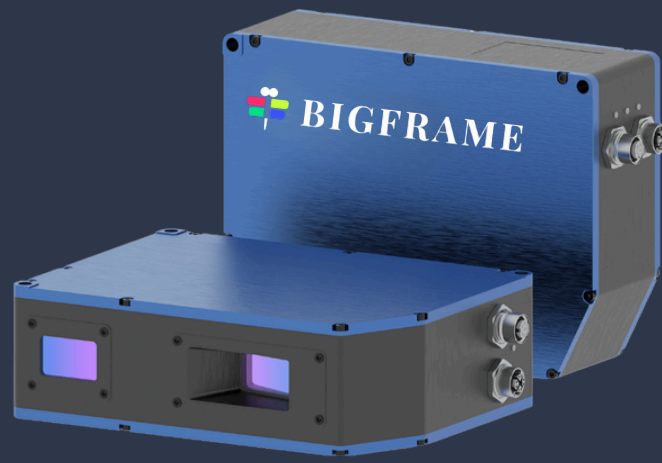


TridiVision-Universal Platform Software

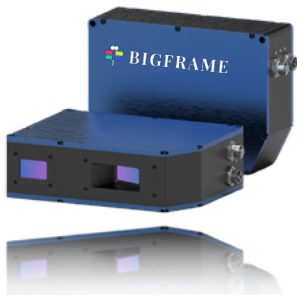
TridiVision

- 2D+3D+AI with 270+ built-in operators
- Deep learning support, plus rapid AI model building from just a few samples
- Ready-made sample programs for typical applications, so you can piece together vision solutions your way
- Covers the full range of demands — positioning guidance, measurement, defect inspection, object recognition, and more



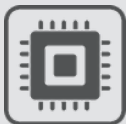


High-accuracy VRH9 Series



Product Description

The high-precision Velociraptor VRH9 series is our proprietary area-scan structured-light 3D camera, with a scan rate up to 6.6 fps and a field of view from 24 mm to 302 mm; built on DLP technology and highly resistant to interference; Z-axis repeatability down to 0.15 μm , suited to high-precision 3D inspection across semiconductor, 3C electronics, and automotive parts.



NVIDIA CUDA



Z-axis repeatability:
0.15 μm



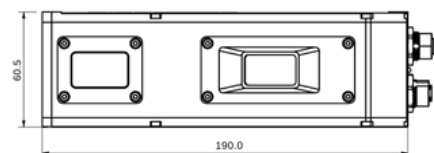
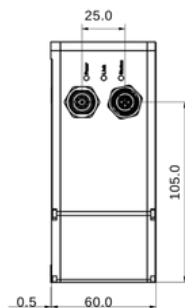
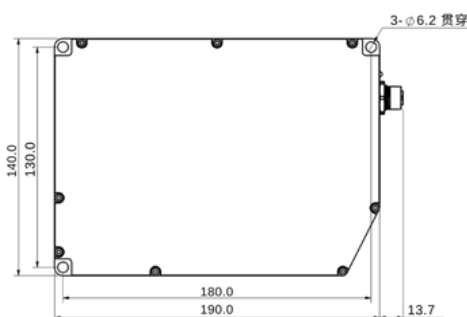
9.07 MP 3D
point cloud



Scan speed:
6.6 fps



FOV:
24 mm – 302 mm

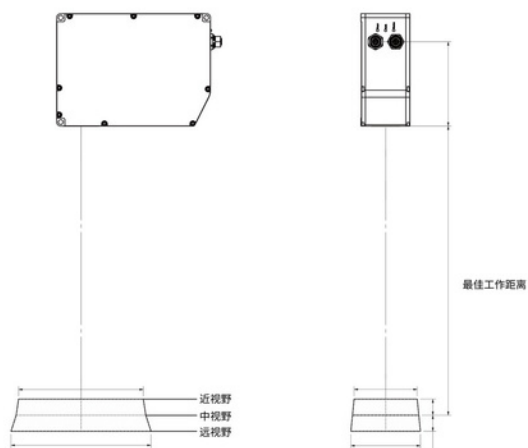


Products Spec

Model	VRH9-020B	VRH9-040B	VRH9-080B	VRH9-120B	VRH9-160B	VRH9-200B	VRH9-280B
Resolution	9.07M 3D Point Cloud						
Scan Range (Z-axis)	112 ~ 114.5 mm	243 ~ 257 mm	260 ~ 280 mm	260 ~ 290 mm	340 ~ 380 mm	330 ~ 390 mm	450 ~ 550 mm
Near-Field (WD, mm)	24.1 x 12.4 mm (112)	47 x 24 mm (243)	79 x 40 mm (260)	120 x 61.5 mm (260)	151 x 77 mm (340)	190 x 98 mm (330)	251 x 129 mm (450)
Mid-Field (WD, mm)	24.3 x 12.5 mm (113)	48.2 x 24.7 mm (250)	81.5 x 41.5 mm (270)	126 x 64.5 mm (275)	159 x 81.5 mm (360)	205 x 105 mm (360)	277 x 142 mm (500)
Far-Field (WD, mm)	24.6 x 12.6 mm (114.5)	49.5 x 25.5 mm (257)	84 x 43 mm (280)	132 x 67.5 mm (290)	167 x 85 mm (380)	220 x 112 mm (390)	302 x 155 mm (550)
Optimal Working Distance	113 mm	250 mm	270 mm	275 mm	360 mm	360 mm	500 mm
Resolution XY-axis	5.8 μ m	11 μ m	19 μ m	30 μ m	38 μ m	49 μ m	66 μ m
Z-axis Regional Repeatability ⁽¹⁾⁽²⁾	0.15 μ m	0.23 μ m	0.36 μ m	0.5 μ m	0.95 μ m	1.3 μ m	1.8 μ m
Optimal Scanning Volume _(mm)	24.3 x 12.5 x 2.5	48.2 x 24.7 x 14	81.5 x 41.5 x 20	126 x 64.5 x 30	159 x 81.5 x 40	205 x 105 x 60	277 x 142 x 100
Scan Rate	1.1 fps @ (4200 x 2160), 6.6 fps @ (640 x 480)						
Power	≥ 75 W						
Input Voltage	DC 24 V						
Wavelength	Blue Light						
Data Interface	Gigabit Ethernet						
Size	190 x 140 x 60 mm						
Weight	1800 g						

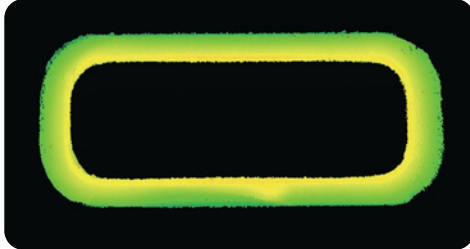
(1) Z-axis Regional Repeatability: The maximum value of the 2-sigma Z-axis repeatability obtained from measurements across 9 evenly divided regions within the Field of View (FOV), when imaging a target ceramic plate at near-field, mid-field, and far-field working distances, respectively.
(2) The 1-sigma standard deviation of 100 consecutive measurements of the difference between the average Z-value of Region A and the average Z-value of Region B. Both Region A and Region B are adjacent, and each has an area less than 1% of the total Field of View (FOV).

(Field of view range diagram)

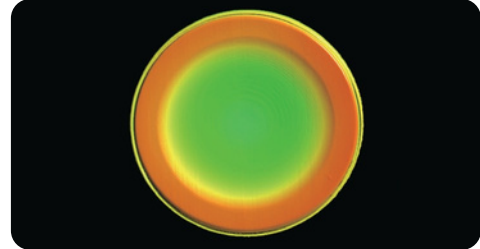


Case Imaging

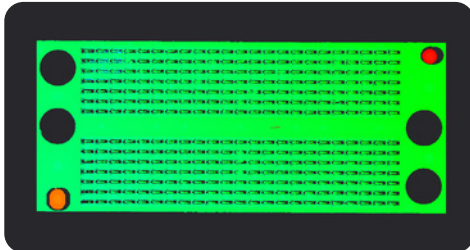
Type-C sealing adhesive



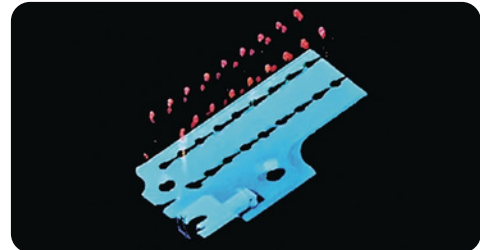
Magnetic charger housing



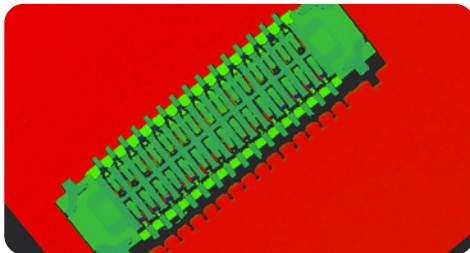
Memory-motherboard connector spring contact



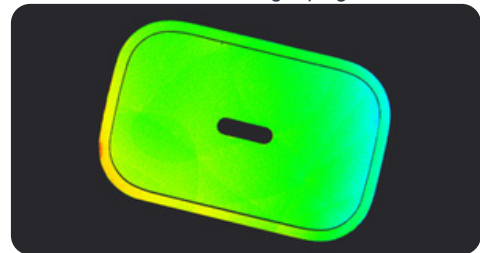
Connector pins



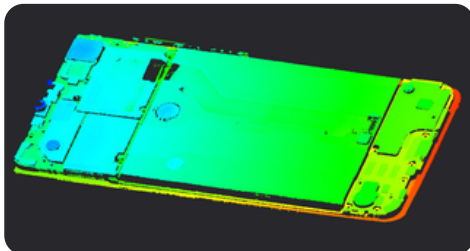
Board-to-board connector



Phone charger plug



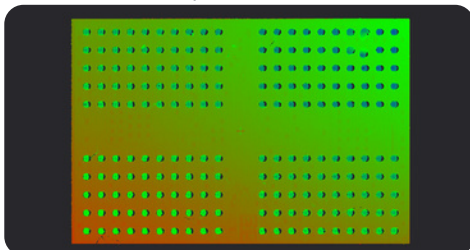
Phone middle frame



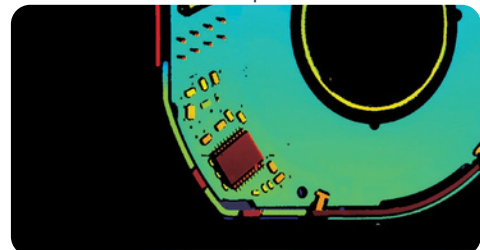
Chip solder balls

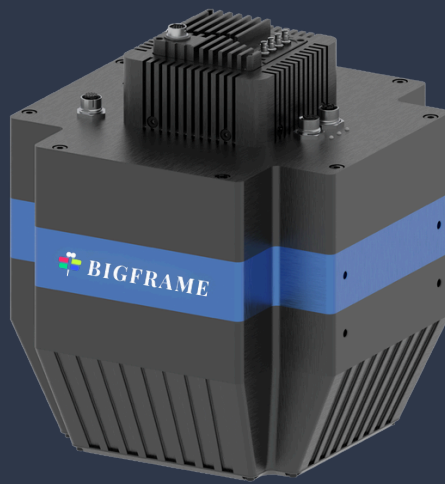


Chip solder balls

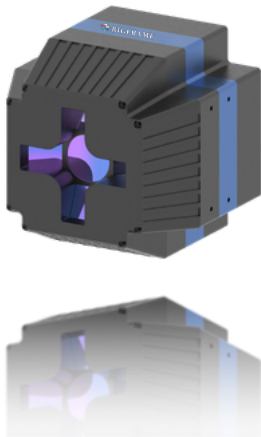


PCB components





Ultra-High Precision Quad-Projection TRS Series



Product Description

The Tyrannosaurus TRS series uses 26 MP 3D point cloud technology, built for precision inspection in the semiconductor and 3C electronics industries. It captures micron-level defects with accuracy and images a full 360° with no blind spots. Powered by a built-in NVIDIA CUDA acceleration engine, it scans in seconds and features HDR imaging for high-quality output under any lighting. New 3D algorithms combined with optimized hardware raise both inspection accuracy and throughput, making it a strong fit for precision inspection.



Semiconductor



Z-axis repeatability:
0.06 μm



26 MP 3D point cloud

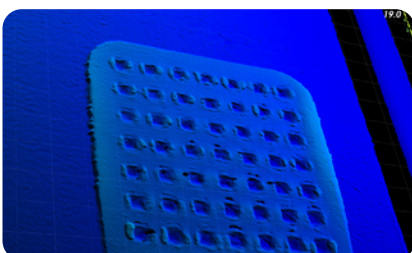


Four-projector

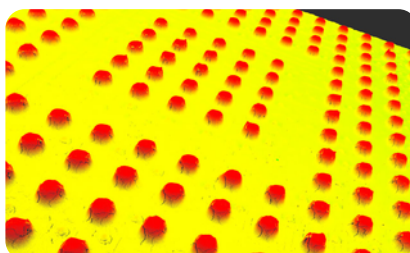


FOV:
9.8 mm – 25.5 mm

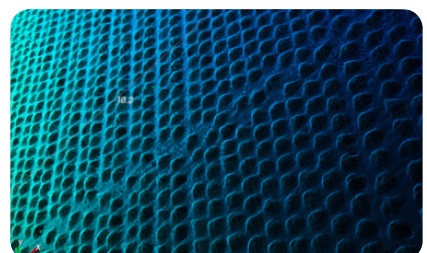
Case Imaging



CCS direct-weld joint AVI inspection



Chip solder balls

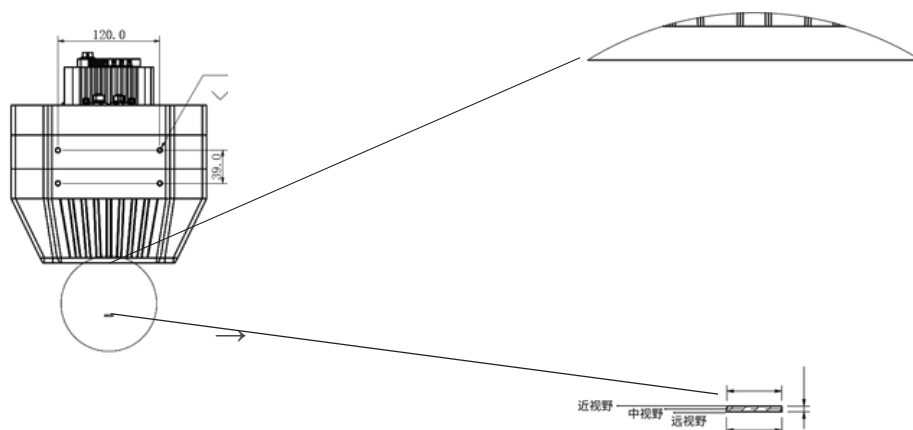


Wafer Bump (wafer pad bump)

Products Spec

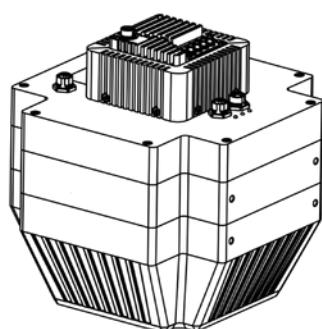
Model	TRS-010B	TRS-025B
Resolution	26M 3D Point Cloud	
Scan Range (Z-axis)	1 mm	5 mm
Near-Field (WD, mm)	9.8 x 9.8 mm (62)	25.5 x 25.5 mm (101.5)
Mid-Field (WD, mm)	9.8 x 9.8 mm (62.5)	25.5 x 25.5 mm (104)
Far-Field (WD, mm)	9.8 x 9.8 mm (63)	25.5 x 25.5 mm (106.5)
Optimal Working Distance	62.5 mm	104 mm
Resolution XY-axis	1.9 μm	5 μm
Z-axis Regional Repeatability ⁽¹⁾⁽²⁾	0.06 μm	0.1 μm
Optimal Scanning Volume	9.8 x 9.8 x 1 mm	25.5 x 25.5 x 5 mm
Scan Rate	$\leq 3 \text{ fps}$	
Power	$\geq 75 \text{ W}$	
Input Voltage	24 V Blue	
Wavelength	Blue Light	
Data Interface	CXP	
Size	230 x 230 x 250 mm	240 x 240 x 275 mm
Weight	8850 g	9740 g

(1) Z-axis Regional Repeatability: The maximum value of the 2-sigma Z-axis repeatability obtained from measurements across 9 evenly divided regions within the Field of View (FOV), when imaging a target ceramic plate at near-field, mid-field, and far-field working distances, respectively.
 (2) The 1-sigma standard deviation of 100 consecutive measurements of the difference between the average Z-value of Region A and the average Z-value of Region B. Both Region A and Region B are adjacent, and each has an area less than 1% of the total Field of View (FOV).

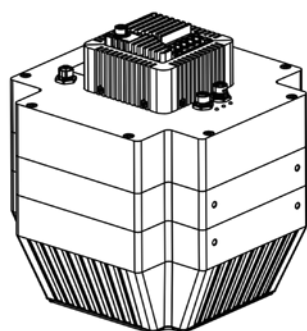
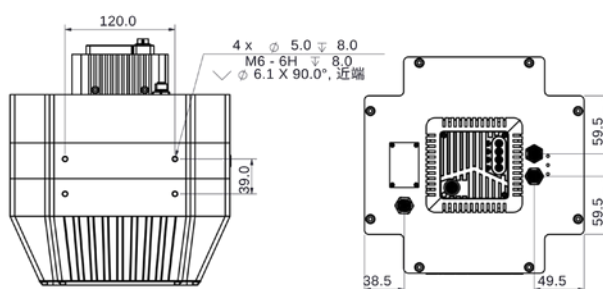
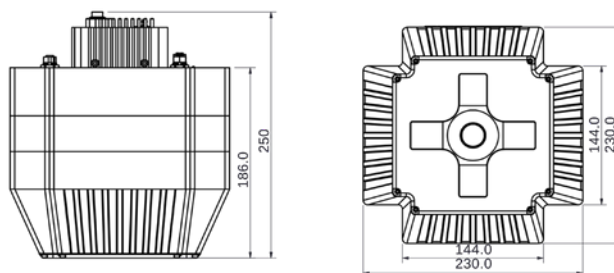


(Field of view range diagram)

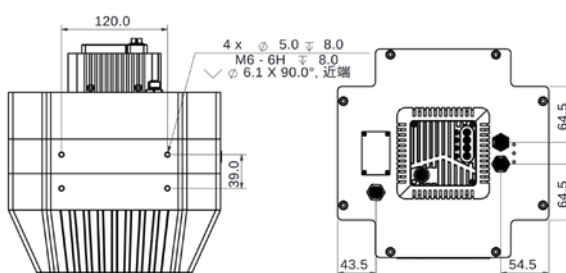
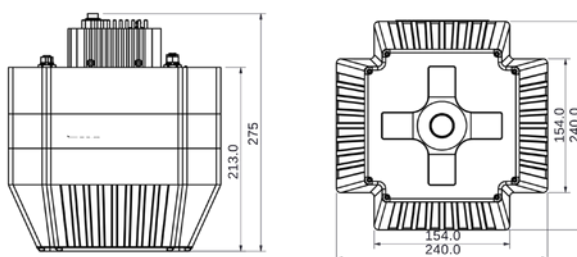
Outside Dimension

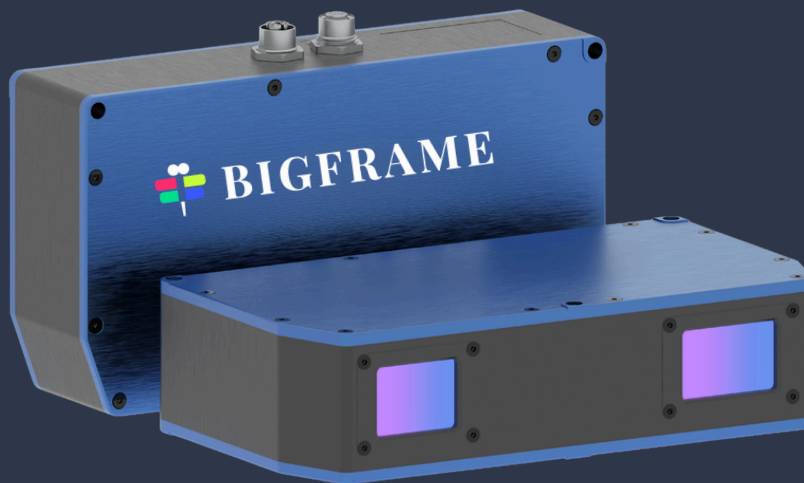


(TRS - 010B)

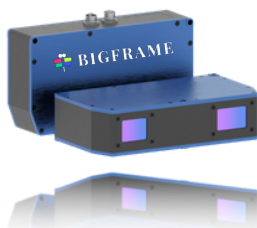


(TRS - 025B)



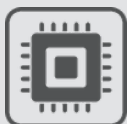


VR Series Area-scanning Structured Light 3D Camera



Product Description

The Velociraptor VR series is our proprietary area-scan structured-light 3D camera, with a field of view from 215 mm to 2180 mm; Z-axis repeatability down to 2 μm and high-quality 3D imaging. Suited to 3D inspection, depalletizing and palletizing, random bin picking, and guidance or positioning.



NVIDIA CUDA



Z-axis repeatability:
2 μm



3.2 MP 3D point cloud

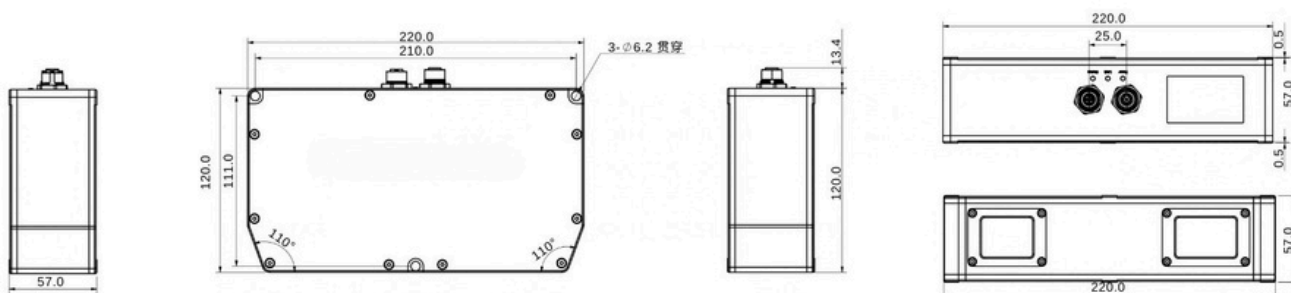


Scan speed:
4 fps



FOV:
215 mm – 2180 mm

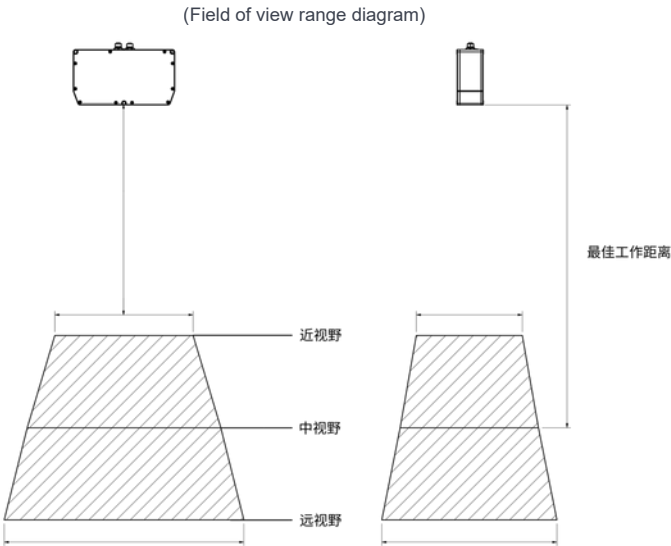
Outside Dimension



Products Spec

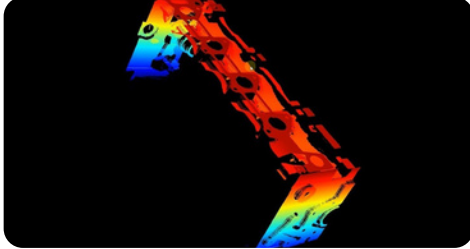
Model	VR-400B	VR-600B	VR-1300B	VR-2300B
Resolution	3.2M 3D Point Cloud			
Scan Range (Z-axis)	350 ~ 650 mm	500 ~ 900 mm	900 ~ 2200 mm	1900 ~ 3800 mm
Near-Field (WD, mm)	215 x 160 mm (350)	300 x 225 mm (500)	530 x 395 mm (900)	1100 x 820 mm (1900)
Mid-Field (WD, mm)	300 x 220 mm (500)	415 x 305 mm (700)	865 x 645 mm (1500)	1500 x 1115 mm (2600)
Far-Field (WD, mm)	380 x 285 mm (650)	525 x 385 mm (900)	1250 x 940 mm (2200)	2180 x 1625 mm (3800)
Optimal Working Distance	500 mm	700 mm	1500 mm	2600 mm
Resolution, XY-axis	0.15 mm	0.20 mm	0.42 mm	0.73 mm
Z-axis Regional Repeatability ^{(1) (2)}	2 μm	6 μm	34 μm	100 μm
Optimal Scanning Volume	300 x 220 x 150 mm	415 x 305 x 200 mm	865 x 645 x 700 mm	1500 x 1115 x 1200 mm
Scan Rate	1.1 fps @ (2048 x 1536), 4 fps @ (640 x 480)			
Power	≥ 75 W			
Input Voltage	DC 24 V			
Wavelength	Blue Light			
Data Interface	Gigabit Ethernet			
Size	220 x 120 x 56 mm			
Weight	1600 g			

(1) Z-axis Regional Repeatability: The maximum value of the 2-sigma Z-axis repeatability obtained from measurements across 9 evenly divided regions within the Field of View (FOV), when imaging a target ceramic plate at near-field, mid-field, and far-field working distances, respectively.
 (2) The 1-sigma standard deviation of 100 consecutive measurements of the difference between the average Z-value of Region A and the average Z-value of Region B. Both Region A and Region B are adjacent, and each has an area less than 1% of the total Field of View (FOV).

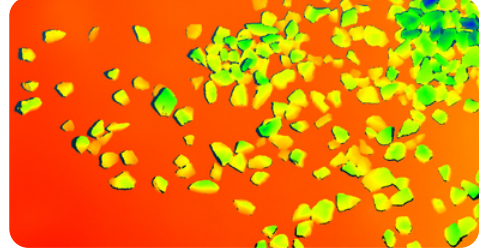


Case Imaging

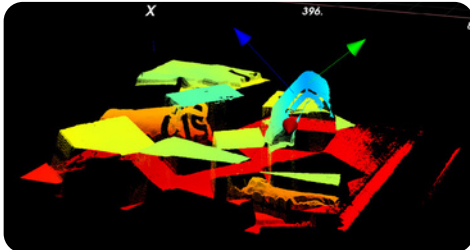
Guided grinding of engine castings



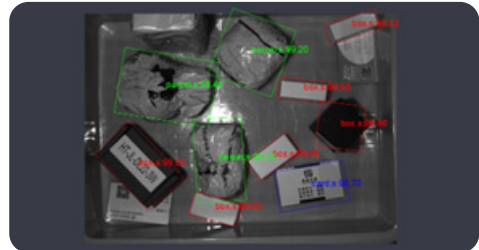
Aggregate size inspection



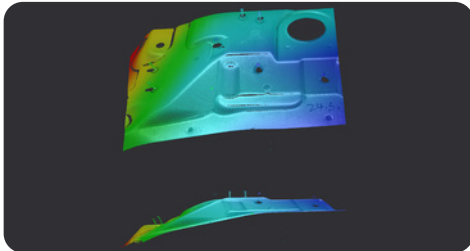
Random bin picking of mixed parcels



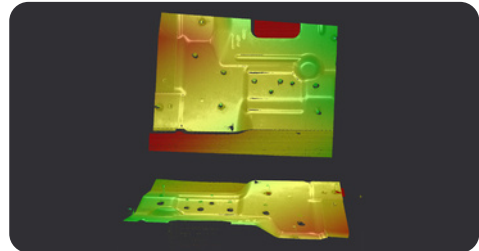
Random bin picking of mixed parcels



Automotive body parts



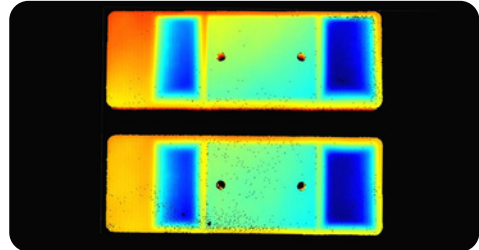
Automotive body parts



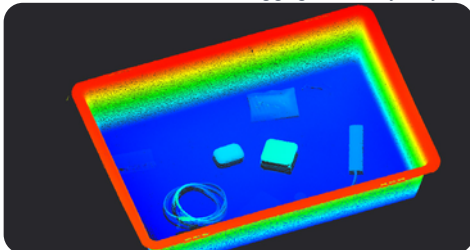
Random picking and guidance of yarn spindles



3D inspection of twin-rail sleepers

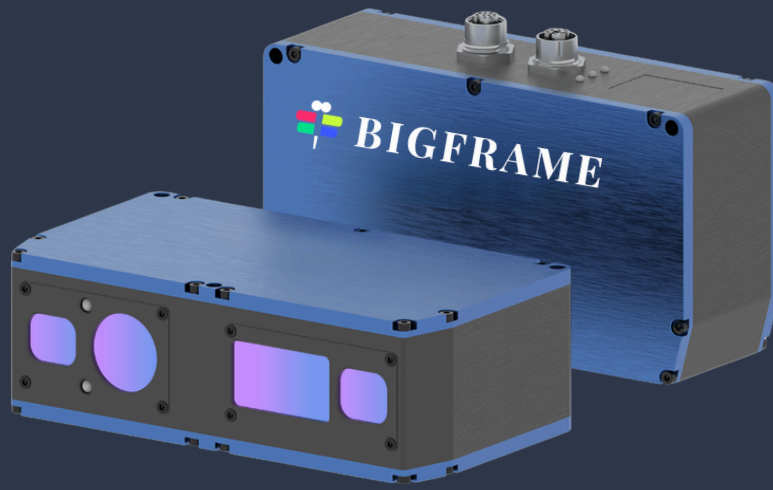


Left-behind items in baggage security trays

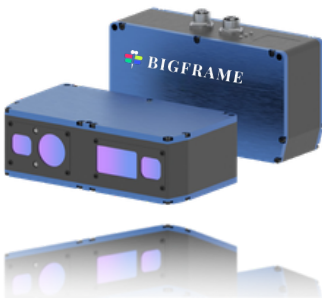


Guided insertion and removal at the filling port





Color Stereo VRD Series



Product Description

The color binocular Velociraptor VRD series pairs 5 MP 3D point cloud technology with DLP structured-light imaging, built for 3D guidance on collaborative robot arms. It outputs RGB color images to raise both surface-inspection and product-segmentation efficiency. Its binocular imaging cuts occlusions and sharpens depth perception, holding up in complex industrial settings. A flexible design meets industrial standards and resists vibration, shock, and ambient light, delivering a stable, efficient cobot-guidance solution for high-dynamic production lines.



Supports RGB image output



Z-axis repeatability:
6 μ m



5 MP 3D point cloud

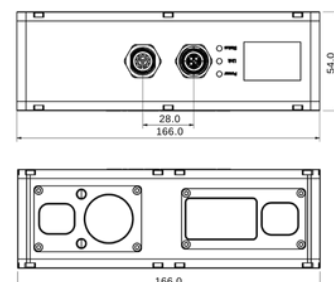
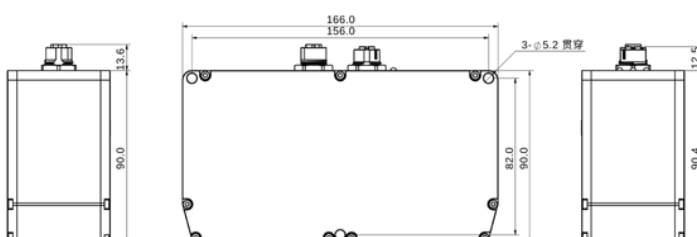


Binocular 3D
imaging, lightweight
& compact



FOV:
365 mm to 1538 mm

Outside Dimension

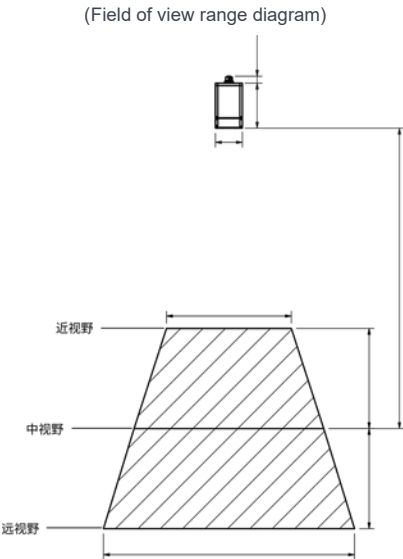


Products Spec

Model	VRD-600B	VRD-1200B
Resolution	5M 3D Point Cloud	
Scan Range (Z-axis)	400 ~ 800 mm	900 ~ 1600 mm
Near-Field (WD, mm)	365 x 248 mm (400)	887 x 551 mm (900)
Mid-Field (WD, mm)	580 x 369 mm (600)	1166 x 731 mm (1200)
Far-Field (WD, mm)	795 x 490 mm (800)	1538 x 974 mm (1600)
Optimal Working Distance	600 mm	1200 mm
Resolution XY-axis	0.24 mm	0.48 mm
Z-axis Regional Repeatability ^{(1) (2)}	6 μm	24 μm
Optimal Scanning Volume	580 x 369 x 300 mm	1166 x 731 x 400 mm
Scan Rate	≤ 1.2 fps	
Power	≥ 75 W	
Input Voltage	DC 24 V	
Wavelength	Blue Light	
Data Interface	Gigabit Ethernet	
Size	166 x 90 x 54 mm	
Weight	870 g	

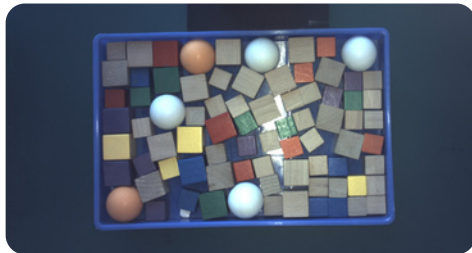
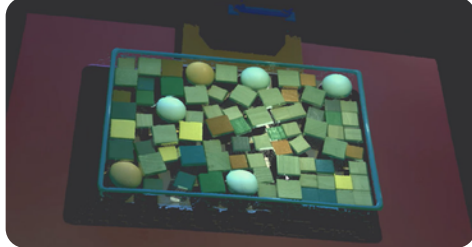
(1) Z-axis Regional Repeatability: The maximum value of the 2-sigma Z-axis repeatability obtained from measurements across 9 evenly divided regions within the Field of View (FOV), when imaging a target ceramic plate at near-field, mid-field, and far-field working distances, respectively.

(2) The 1-sigma standard deviation of 100 consecutive measurements of the difference between the average Z-value of Region A and the average Z-value of Region B. Both Region A and Region B are adjacent, and each has an area less than 1% of the total Field of View (FOV).



Case Imaging

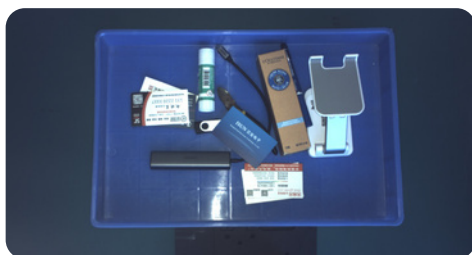
Segmentation and picking of colored blocks



Recognition of common tools



Recognition of left-behind items in luggage trays



Carton picking and surface print-quality inspection



Segmentation and picking of snack packaging



Screw recognition and picking



TRIDIVISION

Universal Software Platform

TridiVision Universal Software Platform

TridiVision software is a powerful graphical development platform designed specifically for machine vision engineers, offering industry-leading functionality and flexibility. It utilizes an intuitive dataflow programming architecture and includes a rich set of powerful image analysis operators, significantly improving development efficiency.

Product Description

TridiVision software is a simple and efficient vision platform that supports 2D/3D vision tasks and multi-brand cameras, allowing you to quickly build various application solutions.

The advantages are as follows:



Integrates over 270+ operators, supporting deep learning and AI rapid modeling with small samples



Meet complex and diverse application requirements in 2D and 3D vision fields, including positioning guidance, measurement, defect detection, and object recognition



Expanded toolset functionality, achieving high modularity and engineering readability



Powerful self-developed contour matching algorithm, high speed, high precision, and supports scaling matching and perspective deformation matching



HMI interface with rich control types



Support for Automatic Robot Path Planning and 3D Trajectory Generation



Deep pseudo-color point cloud acquisition, real-time preview, loop acquisition, repeatability testing readability



Provides sample programs for various typical applications, allowing for flexible development of visual application solutions



Support customized algorithm development

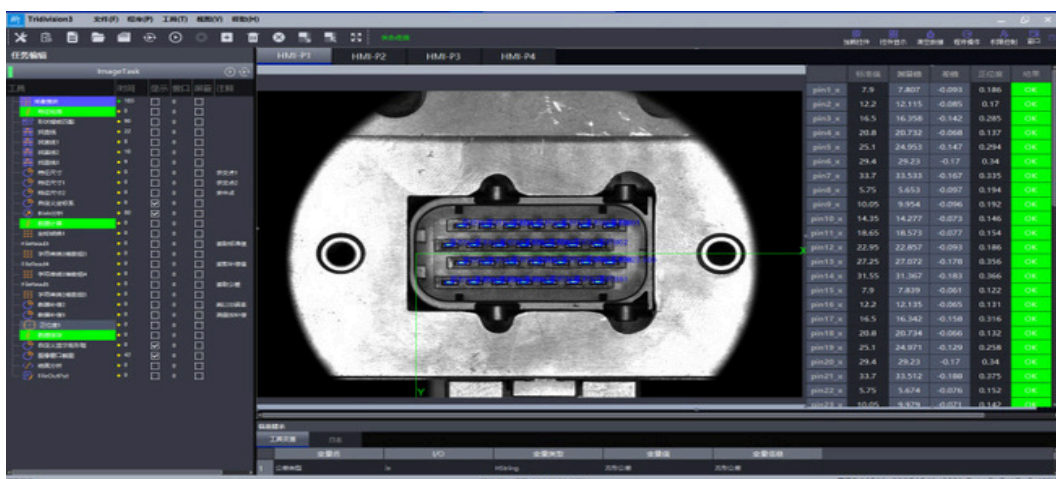
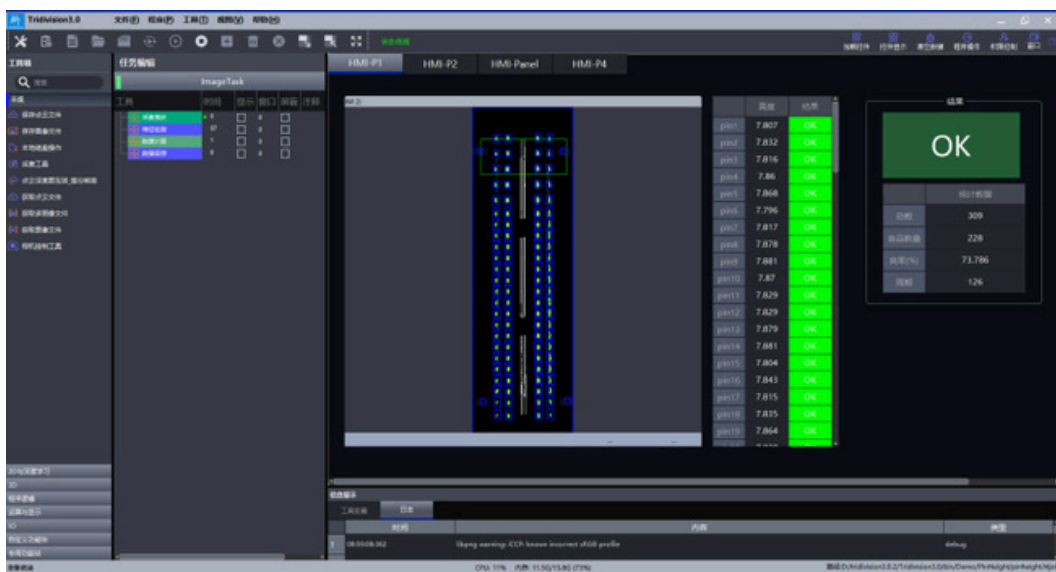
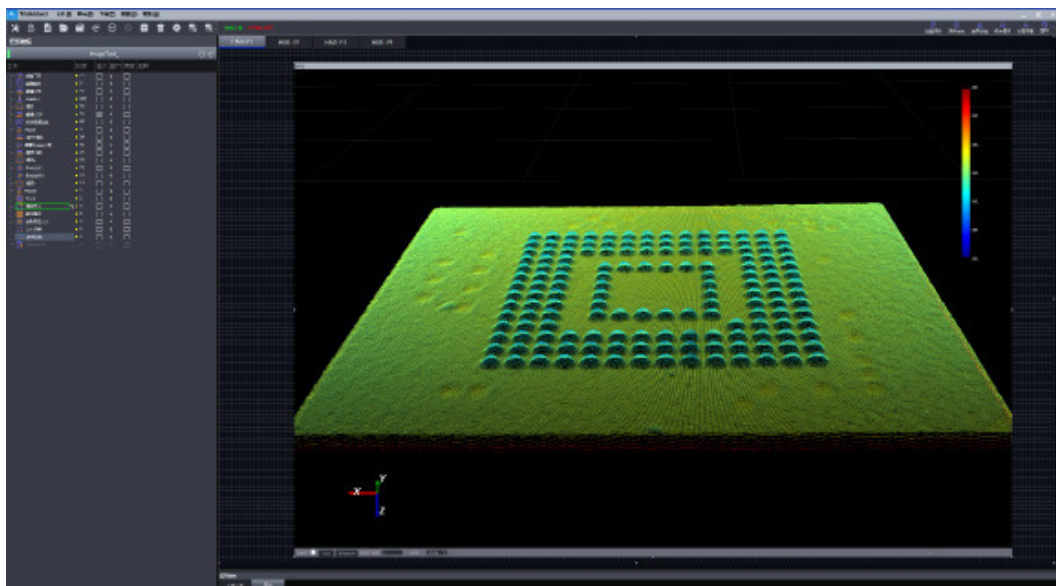


Supports numerous third-party 2D and 3D camera brands, including Hikvision, Basler, Daheng, LMI, Keyence, Smartray, and Deep Vision

Version & Function Description

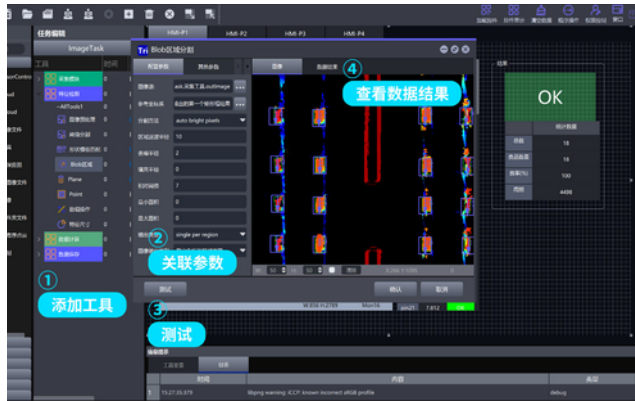
Software Version	Supported Function Modules															
	1D	2D		3D		3D-zmap			Image Processing	Point Cloud Processing	Defect Detection	Deep Learning		Calibration	Operation	Communication
	Detection	Position	Detection	Identify	Path Planning	Position	Detection	Process				Training	Reasoning			
TridiVision v3.0-2D Red	✓	✓	✓						✓		✓			✓	✓	✓
TridiVision v3.0-3D-zmap(2D) Orange	✓	✓	✓			✓	✓	✓	✓		✓			✓	✓	✓
TridiVision v3.0-3D-poincloud(2D+3D-zmap) Yellow	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓			✓	✓	✓
TridiVision v3.0-AI-running(2D+3D-zmap) Green	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓		✓	✓	✓	✓
TridiVision v3.0-AI-training(all in one) Blue	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓
HT-E5 BASIC DONGLE (USB hardware key)	/															

Software Interface



Products Introduction

Simple and Fast Graphical Programming



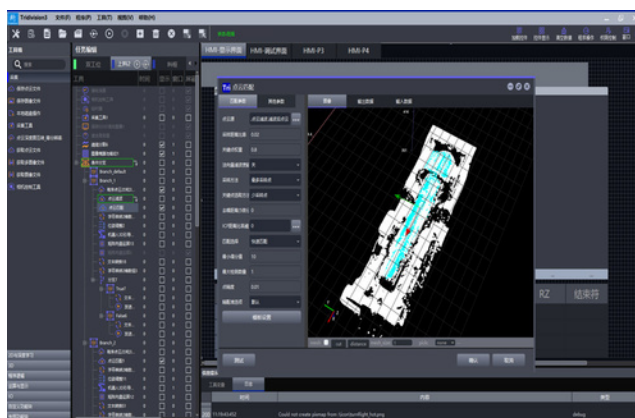
Add tools, associate parameters, test, view data results, and use NO CODE graphical programming for simple programming.

Flexible Human-computer Interaction Interface



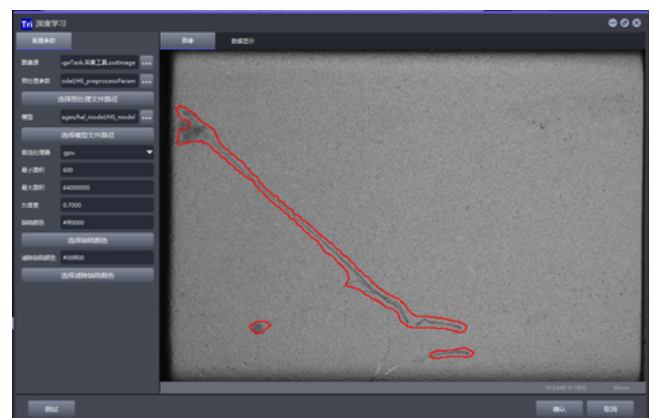
The HMI designer can flexibly drag and drop display controls and input controls to form a beautiful user interaction interface according to user application requirements.

Rich 3D Detection Algorithms



The powerful 3D module supports height measurement, multi-camera stitching, and object recognition with grasping guidance in 3D environments.

Advanced Deep Learning Algorithms



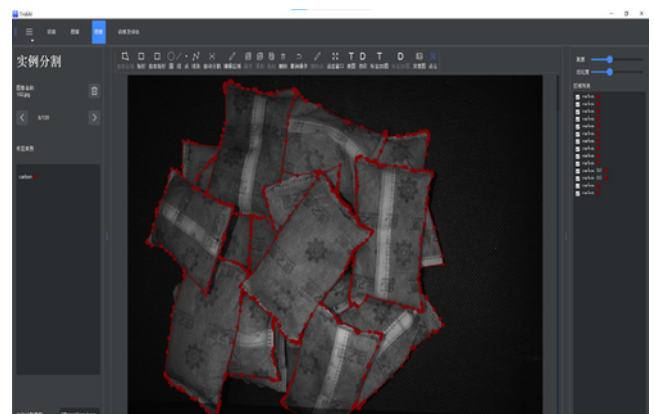
The software's built-in deep learning module supports deep learning classification, semantic segmentation, object recognition and other applications.

Complete Calibration Module



The software has a built-in automatic calibration module, under the guidance of which users can quickly deploy automatic hand-eye calibration in the field of 2D or 3D robot grasping.

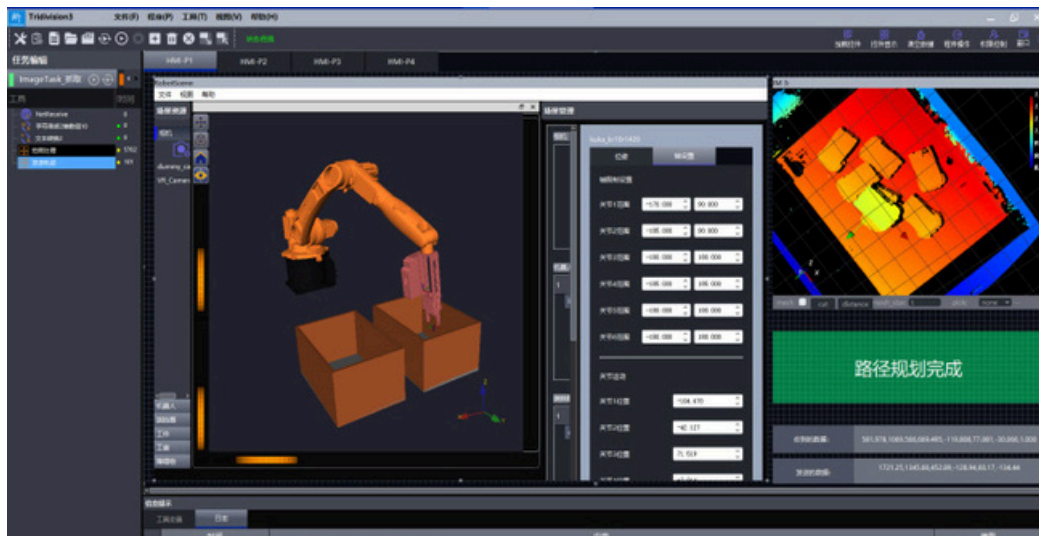
AI Small Sample Rapid Modeling



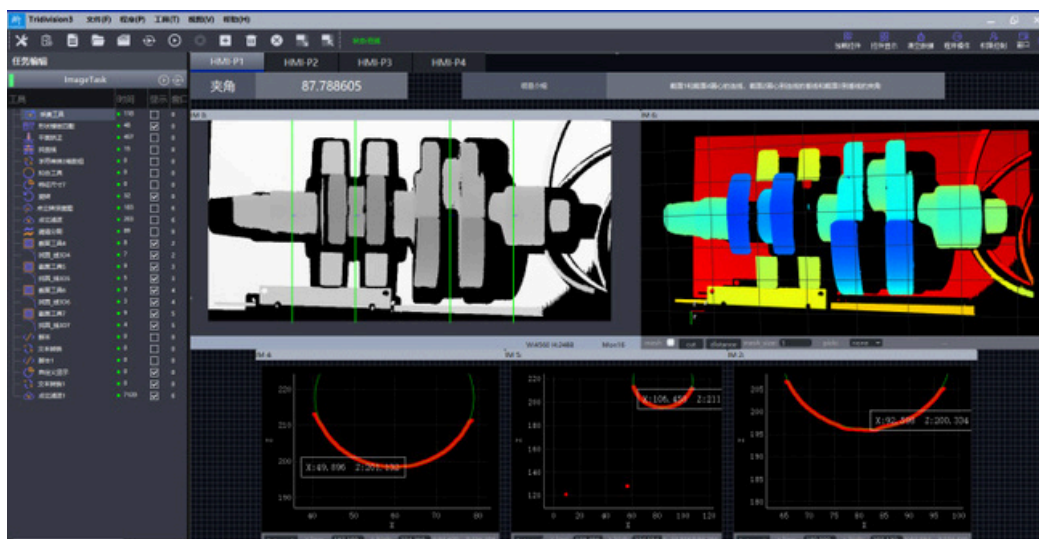
Intelligent simple annotation, small number of image samples, short model training time, support for model optimization and improvement.

Demonstration

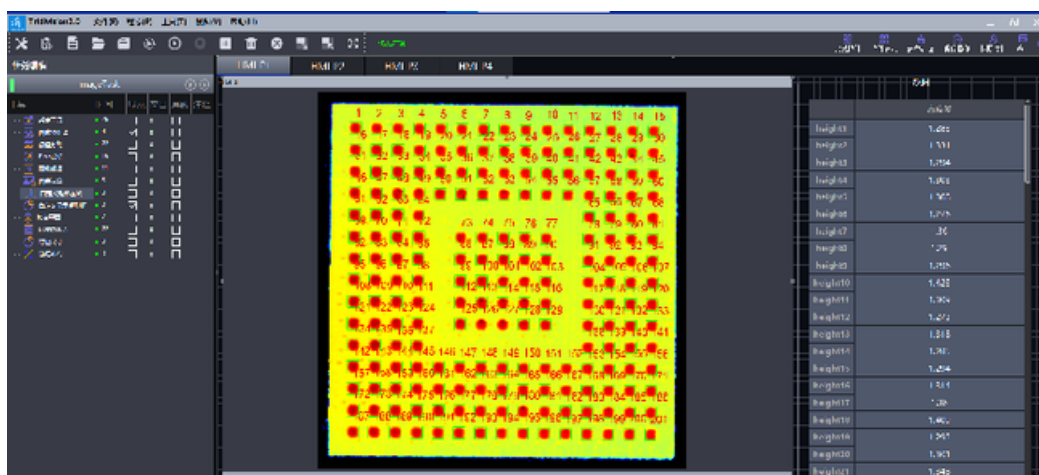
3D Path Planning



3D Height Measurement

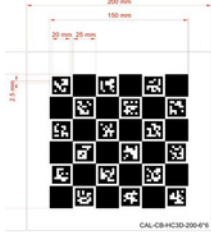
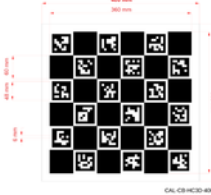
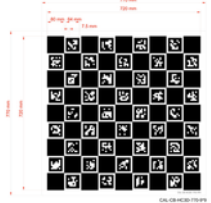
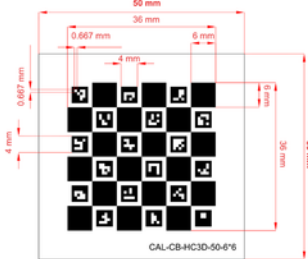
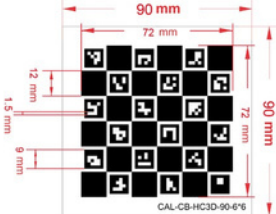
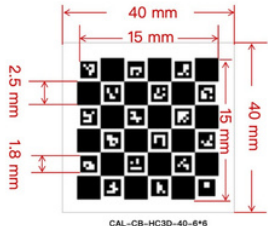


3D Profile Measurement



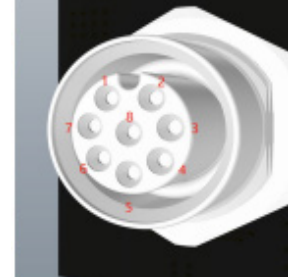
VISION ACCESSORIES OVERVIEW

Calibration Board -3D Camera

Model	Schematic	Accuracy	Overall Dimensions	Array	Corner Points	Grid Size	Pattern Size	Compatible Camera Model
CAL-CB-HC3D-200-6*6		±10μm	200mm x 200mm	6 x 6	5 x 5	25mm x 25mm	20mm x 20mm	VR-400B / VR-600B / VRD-600B
CAL-CB-HC3D-400-6*6		±10μm	400mm x 400mm	6 x 6	5 x 5	60mm x 60mm	48mm x 48mm	VR-1300B / VRD-1200B
CAL-CB-HC3D-770-9*9		±10μm	770mm x 770mm	9 x 9	8 x 8	80mm x 80mm	64mm x 64mm	VR-2300B
CAL-CB-HC3D-50-6*6		±5μm	50mm x 50mm	6 x 6	5 x 5	6mm x 6mm	4mm x 4mm	VRH9-120B / VRH9-160B
CAL-CB-HC3D-90-6*6		±5μm	90mm x 90mm	6 x 6	5 x 5	12mm x 12mm	9mm x 9mm	VRH9-200B / VRH9-280B
CAL-CB-HC3D-40-6*6		±5μm	40mm x 40mm	6 x 6	5 x 5	2.5mm x 2.5mm	1.8mm x 1.8mm	VRH9-040B / VRH9-080B

Cable - 3D Camera

Model	Description	Applicable Products
VR-Power-I/O, 5 m	3D camera power IO cable, 5 m	VR, VRH9 & VRD Series
VR-Power-I/O, 10 m	3D camera power IO cable, 10 m	
VR-Power-I/O, 20 m	3D camera power IO cable, 20 m	
VR-Ethernet Cable, 5 m	3D camera GigE communication cable, 5 m	
VR-Ethernet Cable, 10 m	3D camera GigE communication cable, 10 m	
VR-Ethernet Cable, 20 m	3D camera GigE communication cable, 20 m	
TRS-CP12-HD-HD, 5m	3D camera CXP frame grabber cable, 5 m	TRS Series
TRS-CP12-HD-HD, 10m	3D camera CXP frame grabber cable, 10 m	
TRS-CP12-HD-HD, 20m	3D camera CXP frame grabber cable, 20 m	



Velociraptor 3D Camera Power IO Cable Definition

No	Definition	Description	AWG	Color	
1	Power IN	Power input 24 V	AWG20-22	Pink/Black	
2	GND	Power-input ground	AWG20-22	Blue	
3	GND	Power-input ground	AWG20-22	Blue/Black	
4	IN-	IO input 1 -	AWG23-24	Green/Black	
5	IN+	IO input 1 +	AWG23-24	Green	
6	OUT-	IO input 1 -	AWG23-24	Orange/Black	
7	Power IN	Power input 24 V	AWG20-22	Pink	
8	OUT+	IO input 1 +	AWG23-24	Orange	
9	SHIELD	Chassis shield			

BIGFRAME

ABOUT US

Bigframe Technology Co., Ltd., founded in 2021, is a high-tech enterprise specializing in the R&D and manufacturing of machine vision products. Its wholly-owned subsidiary, Imelligent Technology (Shanghai) Co., Ltd., established in 2015, is a renowned brand in 3D vision, driving technological innovation and industry applications.

Bigframe Technology focuses on developing and promoting standardized 3D industrial vision solutions, addressing key industry challenges and customer needs. Leveraging advanced 3D machine vision technology, the company delivers efficient, precise, and practical intelligent vision solutions, empowering enterprises to achieve smart manufacturing and operational excellence.

Guided by the mission "Empowering machines with intelligent vision, enabling flexible production, and driving industrial smart upgrades," the company promotes widespread adoption of machine vision technology across production lines, factories, and enterprises, accelerating the transition to Industry 4.0.

Committed to craftsmanship, integrity, innovation, and excellence, Bigframe Technology leads industry development through technological innovation, empowering employees, creating value for customers, and driving sustainable progress. The company aspires to be a global leader in machine vision technology, facilitating industrial transformation and contributing to sustainable societal growth.

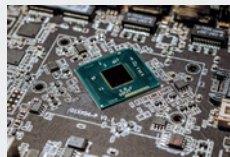
We focus on 3D inspection across industries such as 3C, automotive, semiconductor, lithium battery, and automated inspection, with deep technical expertise in these specialized segments.



3C Manufacturing



New Energy Vehicles



Semiconductor



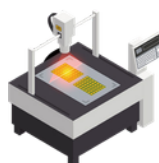
Rail Transit



Automated Factory



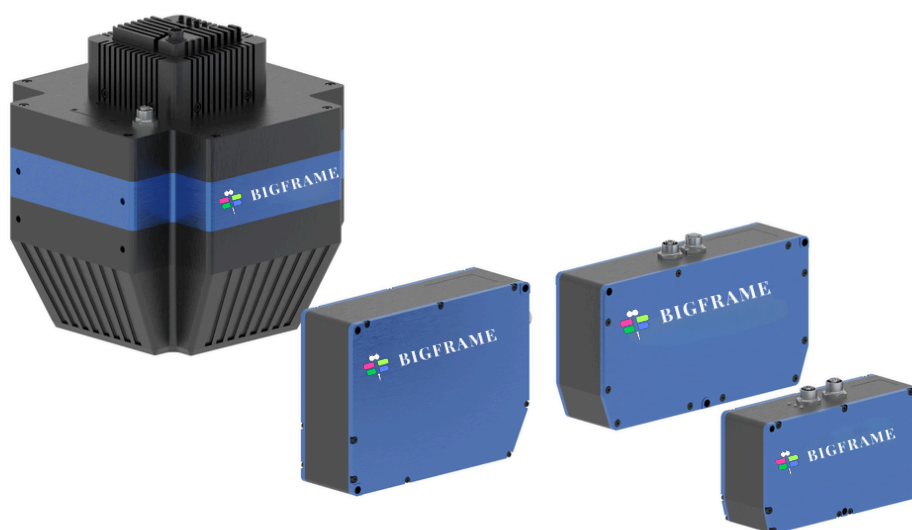
3D Guidance & Positioning



3D Precision Measurement



3D Defect Inspection



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